

1. Goal

The main objective of this assignment is to allow students to explore and apply a chosen data science methodology to a dataset.

2. Assignment preparation

- Select a methodology for the group (2 people)
 - CRISP-DM (Cross-Industry Standard Process for Data Mining)
 - TDSP (Team Data Science Process)
 - SEMMA (Sample, Explore, Modify, Model, Assess)
- Choose a Dataset: Select one of the following datasets (they are available on UCI Machine Learning Repository or Kaggle):
 - **Wine Quality Dataset** (quality of red/white wine based on chemical properties)
 - **Heart Disease Dataset** (Predict whether a person has heart disease based on clinical features)
 - **Titanic** (Predicting survival based on passenger characteristics)
 - **Diabetes Dataset** (predict based on diagnostic measurements whether a patient has diabetes)
 - **Student Performance Dataset** (predict student grades or identify factors affecting performance)
 - **Mall Customer Segmentation Dataset** (Cluster customers based on spending score, income, etc.)
 - **Advertising Dataset** (Predict product sales from advertising spend on TV, radio, and newspaper.)
 - **Penguins Dataset** (Predict penguin species based on flipper length, bill length, etc.)
 - **Rain Prediction of today and tomorrow** (Predict whether it will rain tomorrow based on daily weather reports.)
- **Define Project Goals:** Identify the main goal for your project using the chosen dataset and methodology. Define a central question or objective, such as predicting an outcome (e.g., house prices, flower species, survival chances) or uncovering insights.
- **Apply the Methodology:** Follow the stages of your chosen methodology, documenting your work at each stage.
- **Note:** the combination of methodology/dataset is available at most to one group.
- Fill the tab with the elements of the group, methodology and dataset.

3. Description / Deliverables

The submission consists of a presentation in PPT format and a short report. The presentation will be during class discussion.

The short report can have a front page plus 5 pages (maximum pages allowed). The parts of the report are:

1. Presentation of the framework - e.g present the methodology. You should focus on the stages of the methodologies, advantages, and limitations. Additionally, you should describe how its structure fits the project (≈ 1 page)
2. Stage Summary: List and briefly explain the stages of the chosen methodology (CRISP-DM, TDSP or SEMMA) and how they apply to your project. Try to describe what you would do on each stage ($\approx 1/2$ to 1 page).
3. Implementation of the methodology stages: Complete each stage of your chosen methodology, applying it to the selected dataset. Document your process, decisions and findings for each stage. This includes data exploration and modeling (exploratory data analysis, model selection and evaluation). (1 to 2 pages)
4. A personal analysis of the methodology, the overall findings, including insights, trends or predictions derived from the analysis. Additionally, you can propose recommendation based on your findings. ($\approx 1/2$ page)
5. ... as well as any other relevant insights from the project.

Powerpoint and Short Report Templates are available under Files -> Templates.

Presentation may be in English or in Portuguese.

The presentation will last 10 minutes + 2 minutes for discussion.

4. Group Constitution

Each team will be of 2 people.

5. Peer assessment

You shall question your colleagues during discussion. It will be given extra points for students that participate in QA discussion.

6. References

1. <https://www.datascience-pm.com/crisp-dm-2/>
2. <https://www.datascience-pm.com/crisp-dm-still-most-popular/>
3. <https://www.datascience-pm.com/semma/>
4. Other publicly available sources!